



Hochschule München

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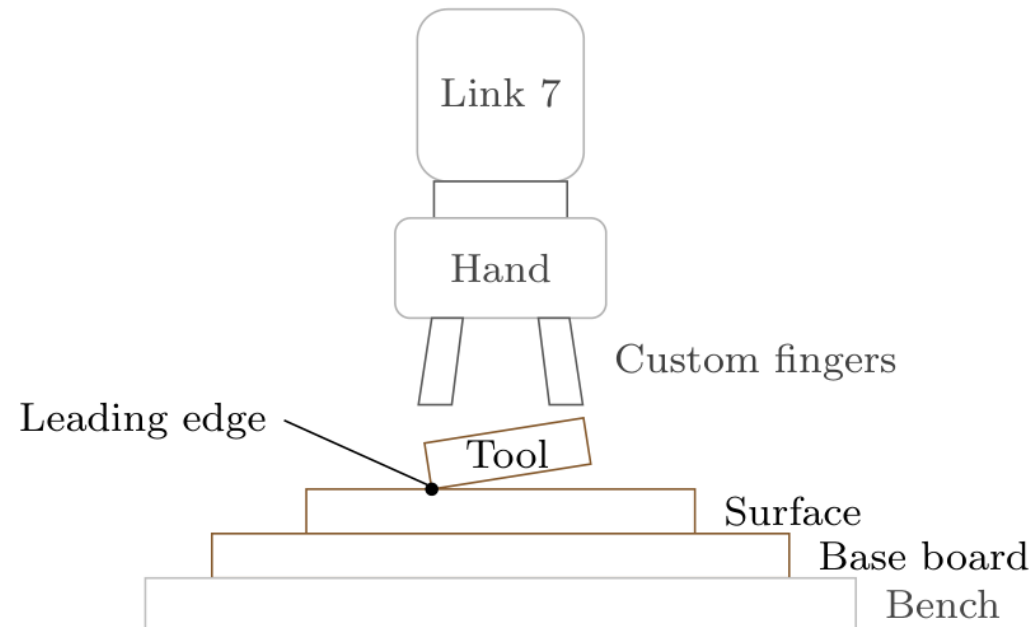
Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

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Problem

- Tilted tool → edge-first contact
- Rigid tracking resists corrective rotation

Main thesis question

- **How do stiffness and the compliance centre affect passive alignment?**

Idea

- Apply the press; let contact rotate the tool

1. Cartesian impedance wrench
2. Real-time control
3. Surface-relative compliance
4. Centre of compliance (CoC)
5. Experimental procedure
6. Angular quantities
7. Baseline contact response
8. Effect of stiffness
9. Effect of CoC position
10. Contact response and interaction moment
11. Secondary study: null-space motion
12. Conclusion

Cartesian wrench = force and moment together

Position and force

Desired / measured p_d, p_{EE}

$$e_p = p_d - p_{EE}$$

$$f = K_p e_p + D_p (\dot{p}_d - \dot{p}_{EE})$$

K_p, K_R translational / rotational stiffness

$\dot{p}, \omega$ linear / angular velocity

Orientation and moment

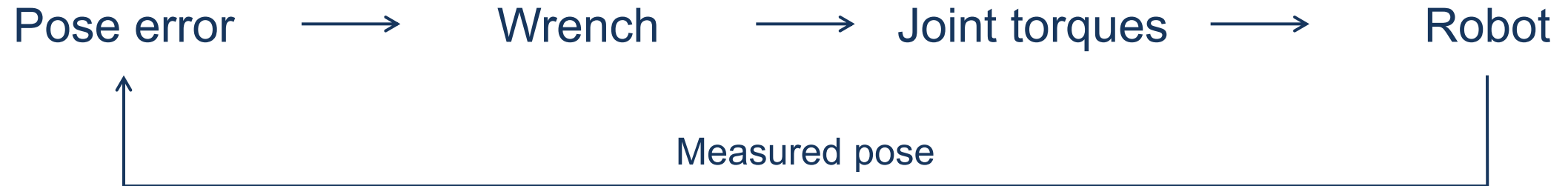
Desired / measured R_d, R_{EE}

e_R rotational correction

$$m = K_R e_R + D_R (\omega_d - \omega_{EE})$$

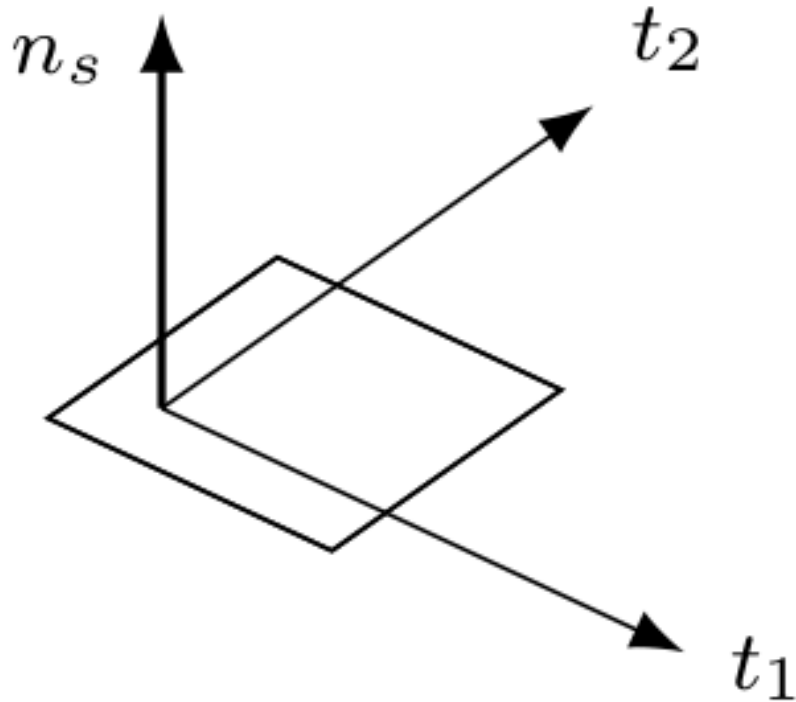
D_p, D_R translational / rotational damping

d, EE desired / measured end effector



$$F = \begin{bmatrix} f \\ m \end{bmatrix}, \quad \tau_{\text{cart}} = J^{\top}(q)F$$

1 kHz control loop · C++ / libfranka



- n_s outward surface normal
- t_1, t_2 two perpendicular surface tangents

Translation

- Softer normal pressing
- Stiffer motion along the surface

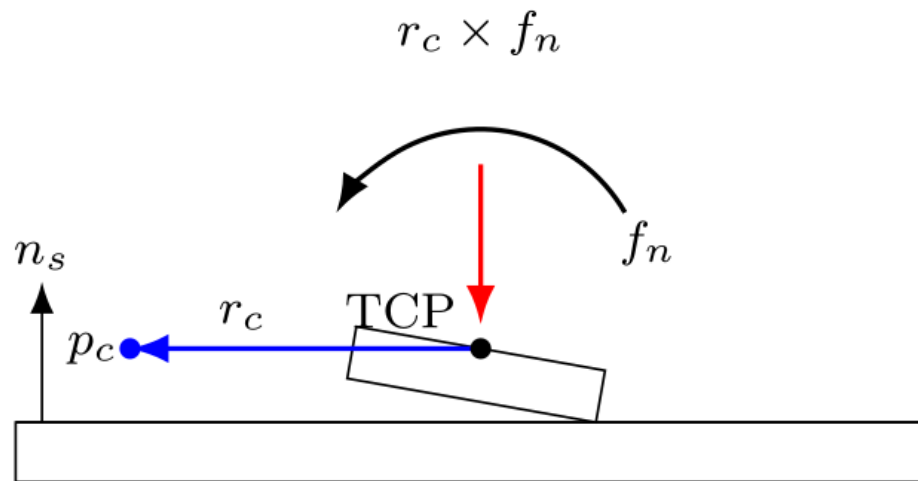
Rotation

- Softer rotations about the tangents
- Stiffer rotation about the normal

Virtual reference point

p_c CoC position

r_c Displacement from TCP to CoC

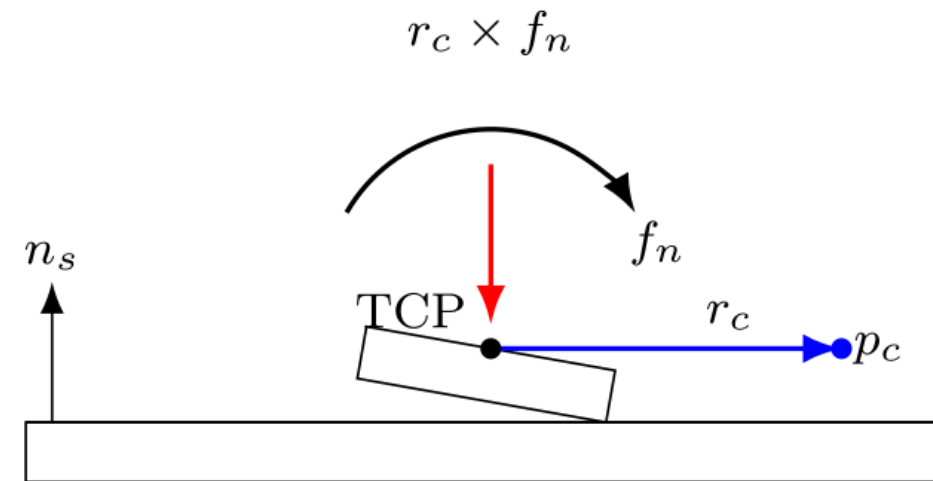


(a) Supporting displacement

Commanded TCP moment

$$m = m_R + r_c \times f$$

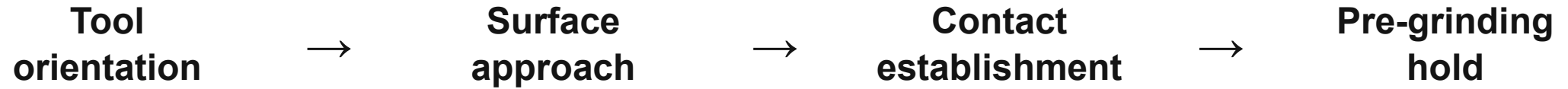
m_R rotational spring-damper moment



(b) Reversed displacement

- Opposite displacement reverses the added moment

Experimental procedure

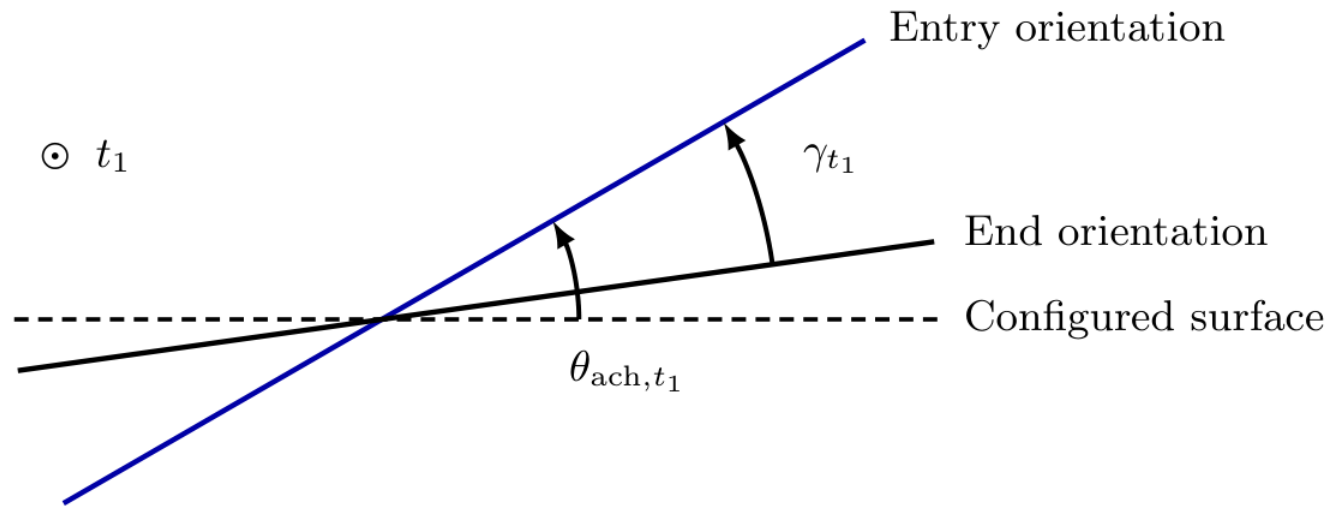


Contact study

- Rectangular tool: 40 × 120 mm
- Five-second contact phase
- Fixed entry orientation reference

Controlled comparison

- A: centre at the TCP
- B / C: rotational / translational stiffness
- D: tangential compliance-centre position
- 57 trials; three repetitions per setting

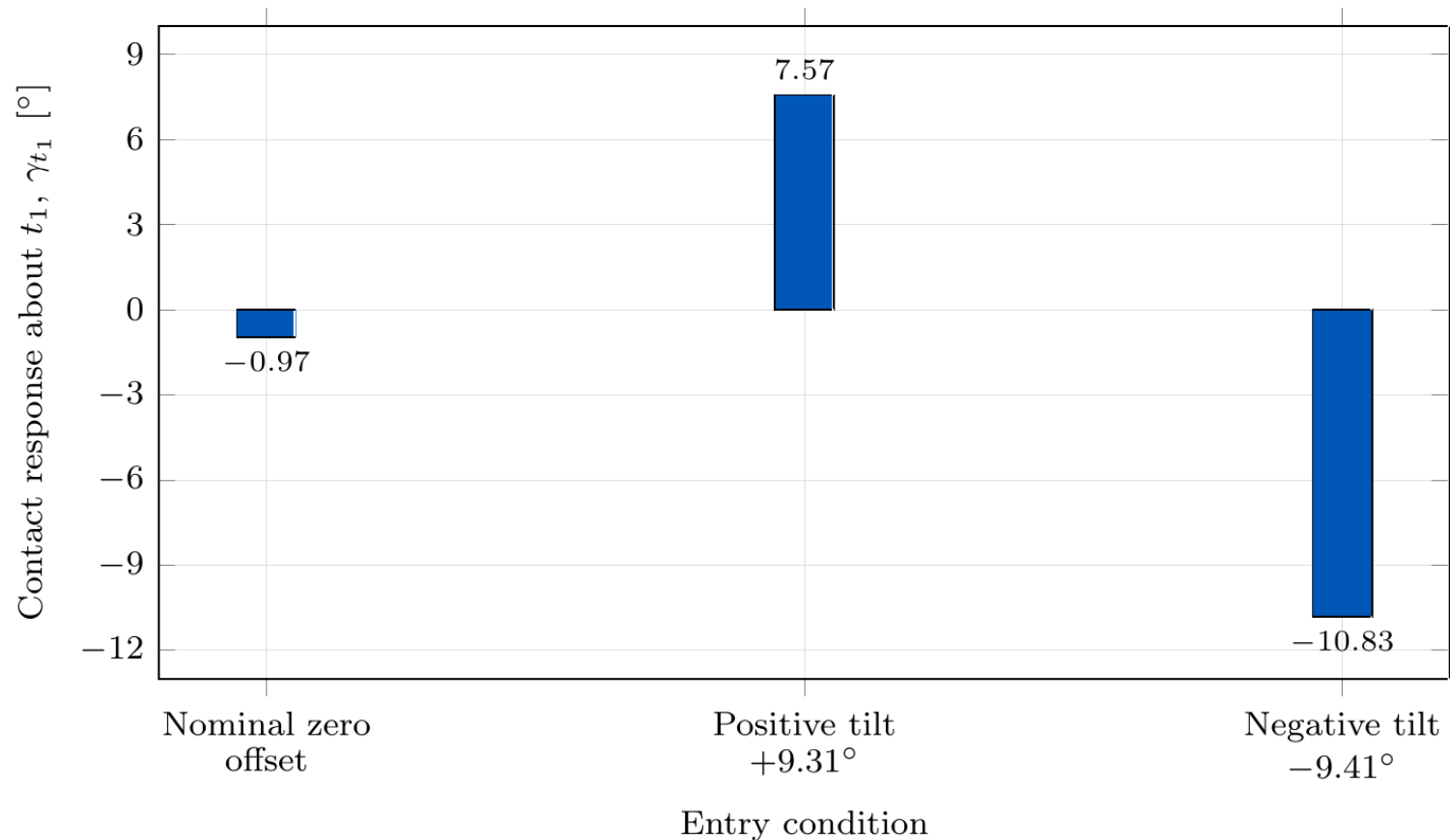


θ_{ach,t_1} Achieved tilt
Orientation at contact entry

γ_{t_1} Contact response
Rotation relative to the held
entry orientation

- Aligning response: same sign as the entry tilt
- Calculated from measured end-effector orientation

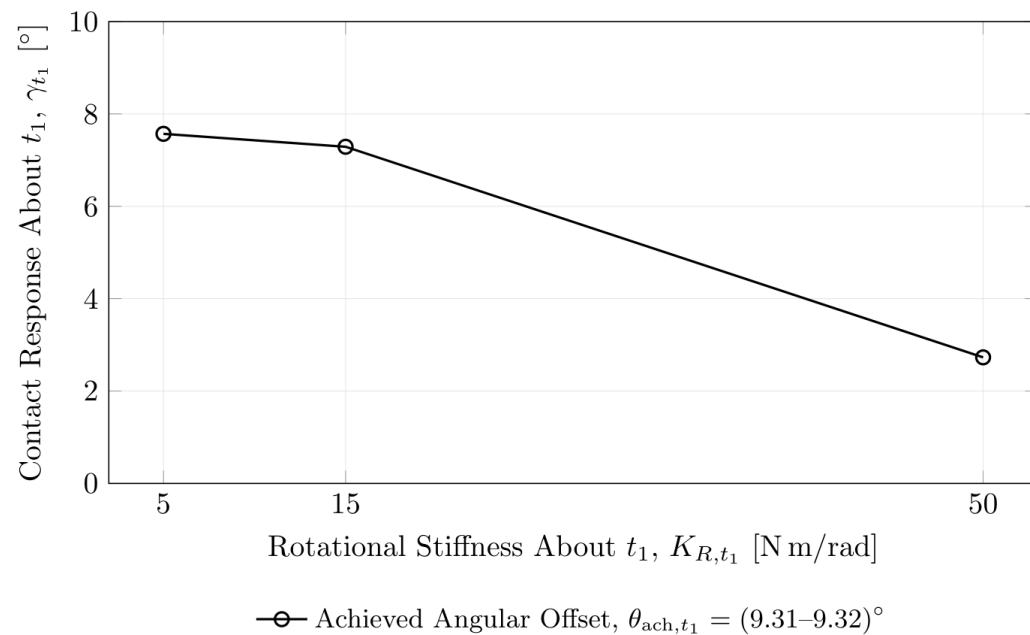
Baseline contact response



- Centre of compliance at the TCP
- Alignment response for both tilt signs
- Unequal response magnitudes

Rotational stiffness

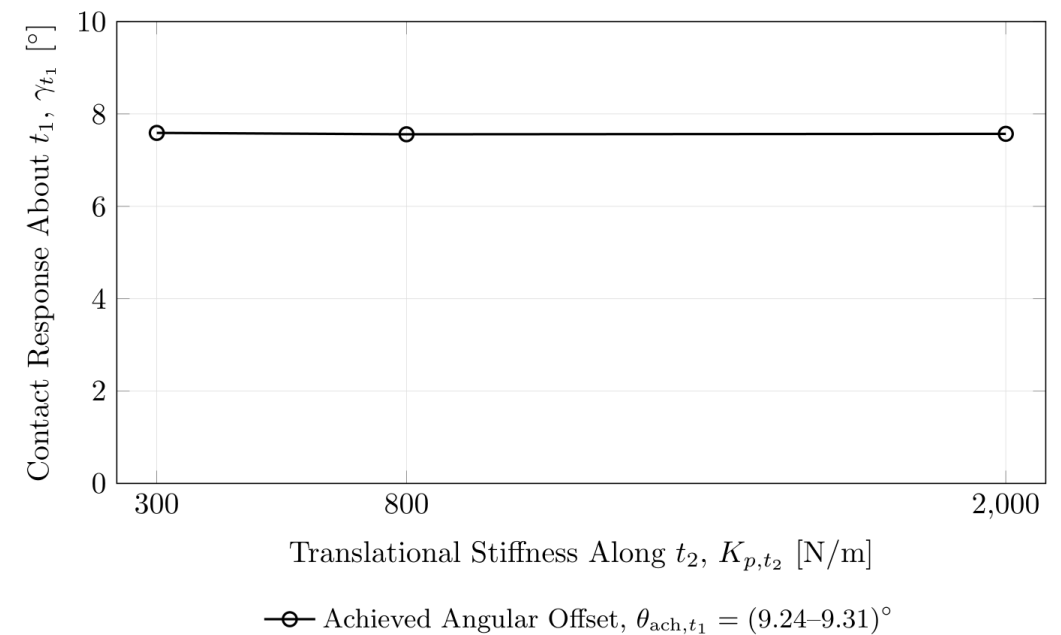
Resistance to rotation about t_1



- Rotational stiffness: 63.9% lower response

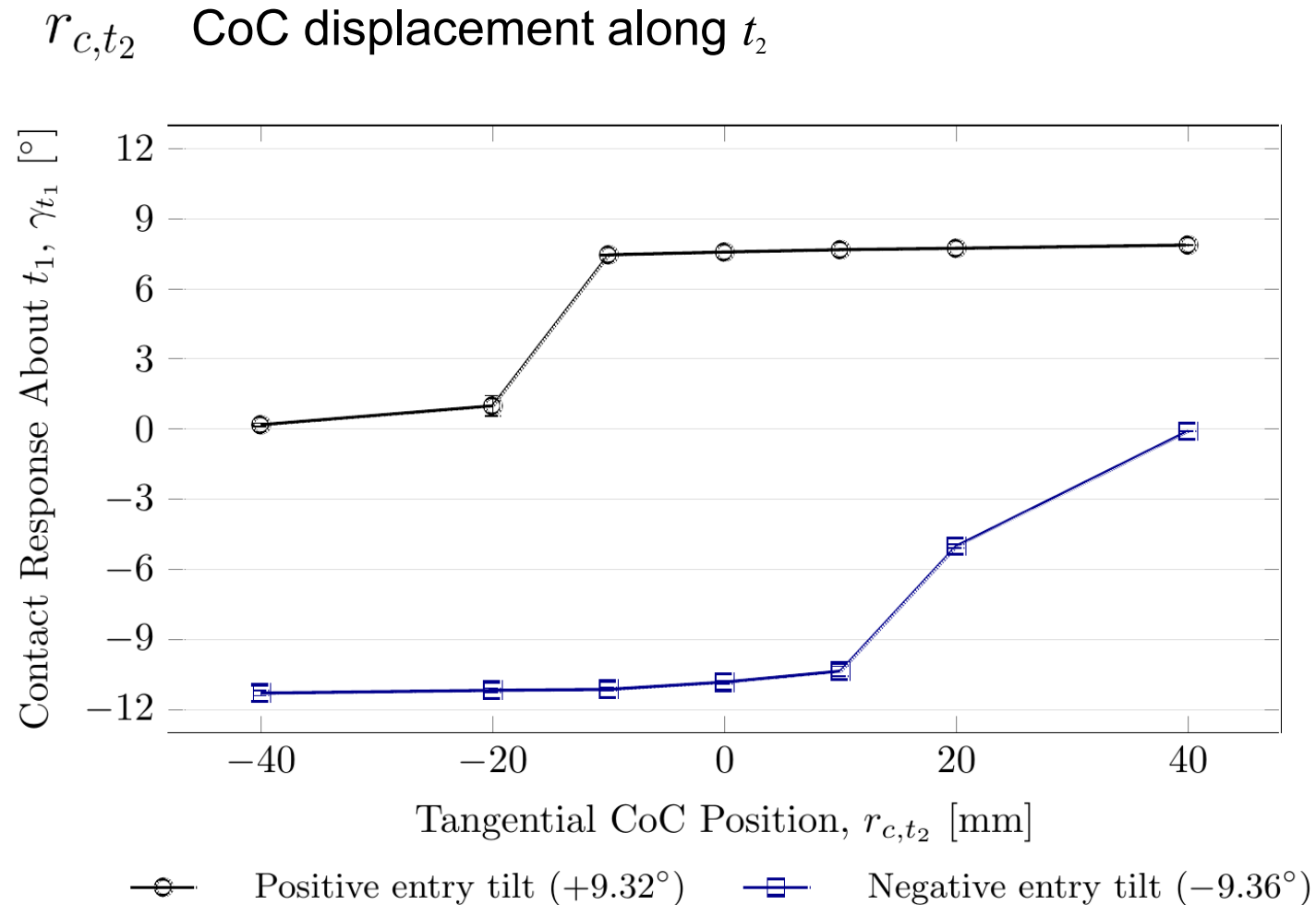
Translational stiffness

Resistance to sliding along t_2



- Tangential stiffness: 0.4% response span

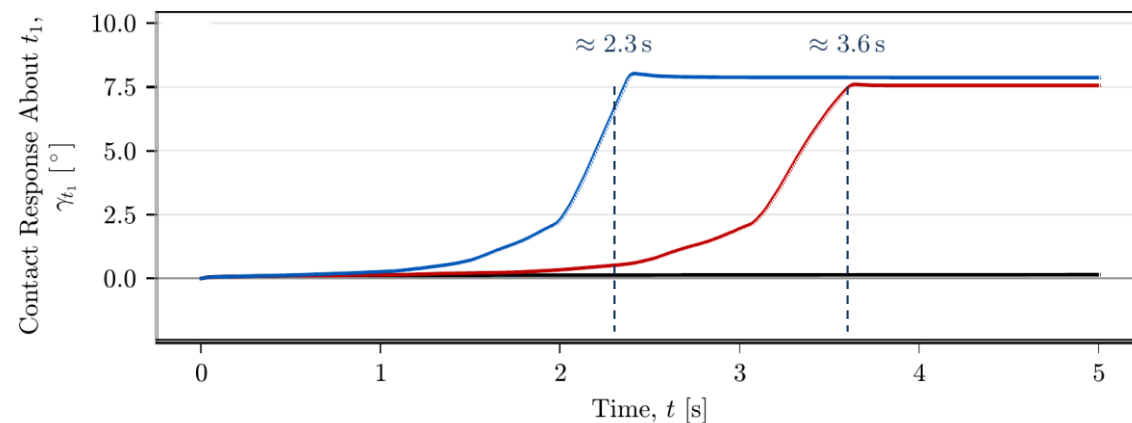
Effect of CoC position



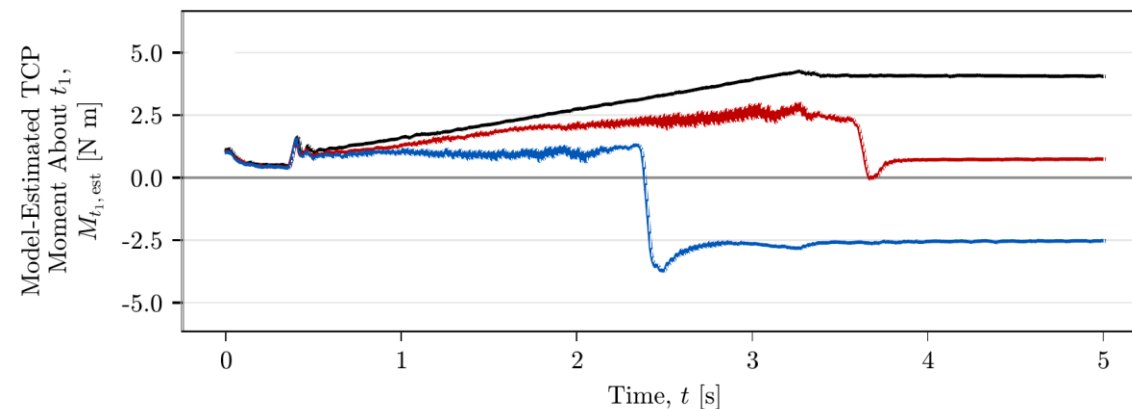
Relative to the TCP-centred baseline

- Supporting displacement: about 4% higher response
- Opposite side: 98–99% lower response
- Required side reverses with the entry tilt

Contact response



Interaction moment at the TCP



— CoC Position, $r_{c,t_2} = -40$ mm — CoC at TCP, $r_{c,t_2} = 0$ — CoC Position, $r_{c,t_2} = 40$ mm

- Reversing CoC displacement reverses the interaction moment
- Supporting displacement: about 1.3 s earlier in these representative traces
- Final normal force (approx.): -83 N (-40 mm), -79 N (TCP), -78 N ($+40$ mm)



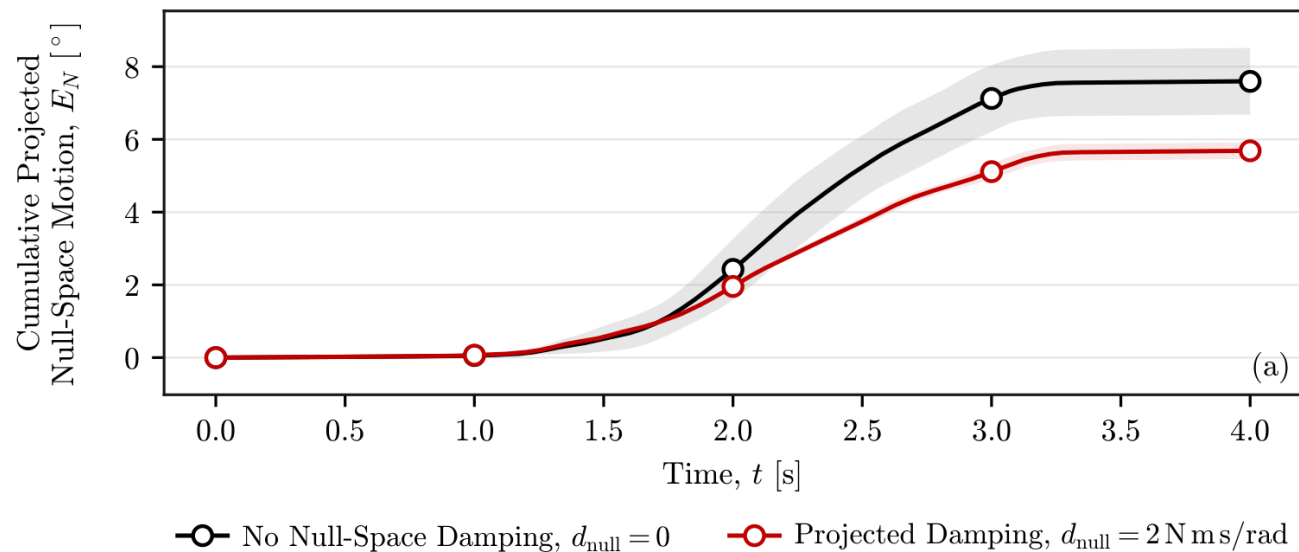
Pre-grinding hold · fixed desired orientation · CoC at TCP

Secondary study: null-space motion

7 joints, 6-DOF tool pose: coordinated motion with the pose held

Damping

Accumulated projected motion



- 25% less accumulated motion
- Maximum TCP position error: 1.304 mm (criterion: 2 mm)

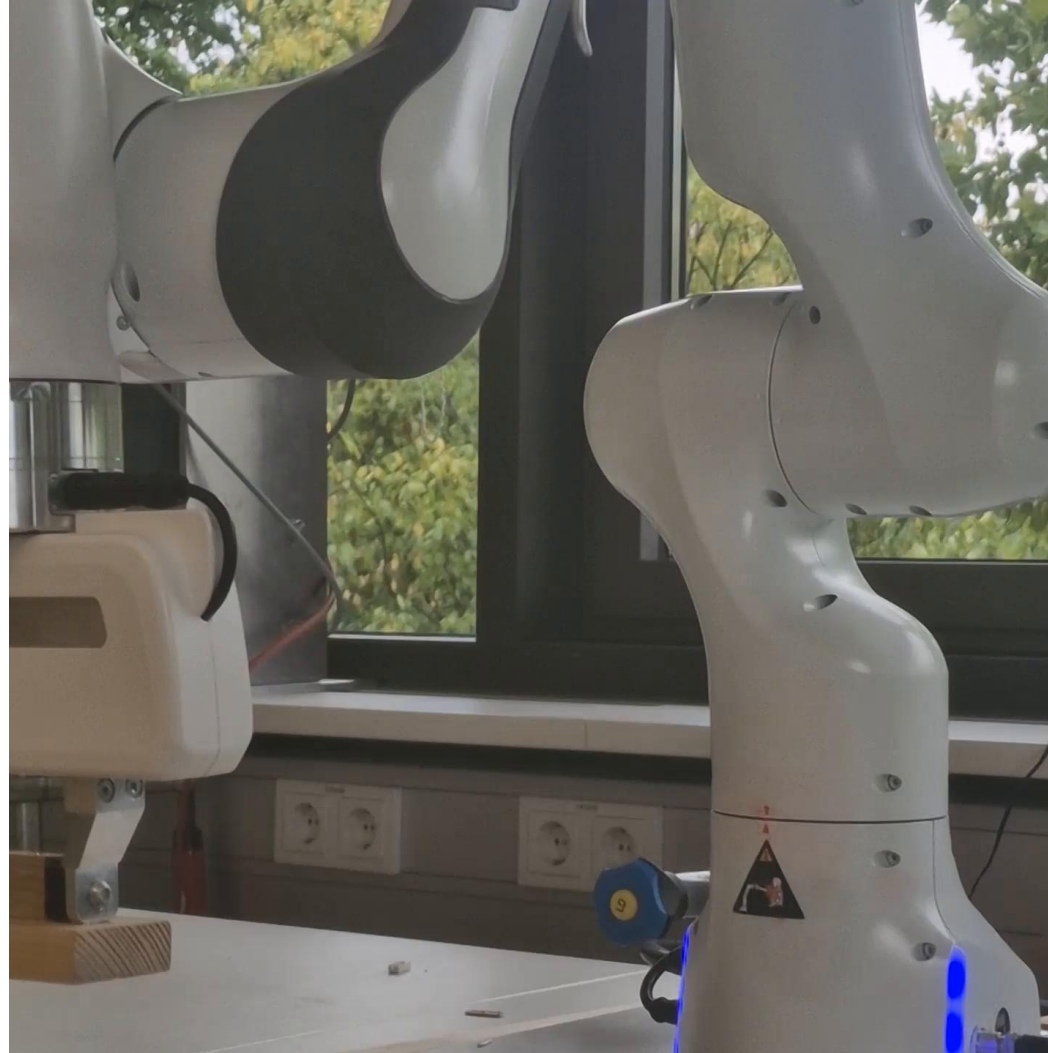
Conditioning

Mean net projected displacement

Setting	Mean (°)
No null-space torque	7.517
Damping	5.609
Conditioning, gain 1.5	0.015
Conditioning, gain 2	-0.006

- Near-zero net displacement
- Stronger gain: more back-and-forth motion

Disturbance demonstration



No null-space torque → Damping → Conditioning

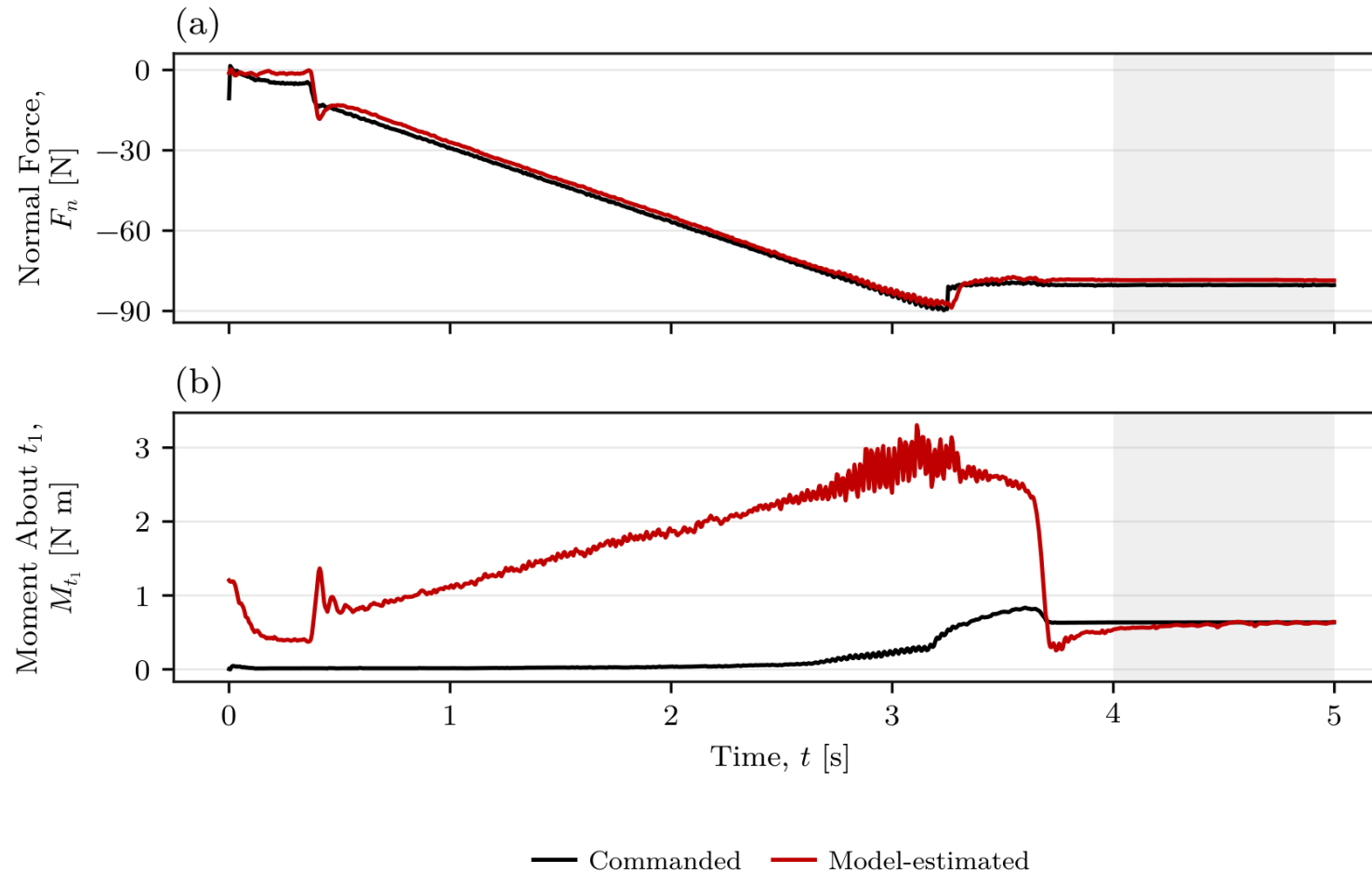
Conclusion

- Passive alignment through Cartesian impedance
- Rotational stiffness controls response magnitude
- CoC displacement selects a preferred rotation direction

Future work

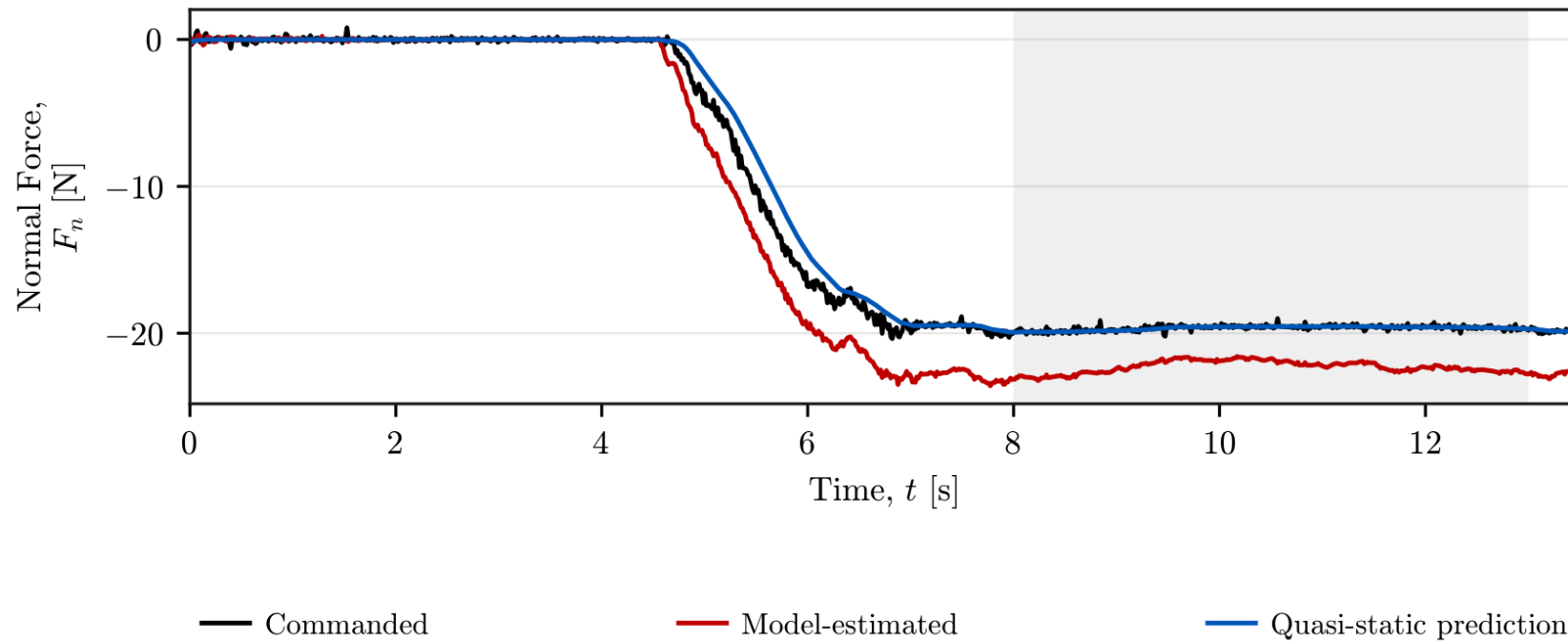
- Tilt-dependent CoC selection
- Return to the TCP after alignment
- Direct angle measurement and sustained grinding

Commanded and estimated wrench



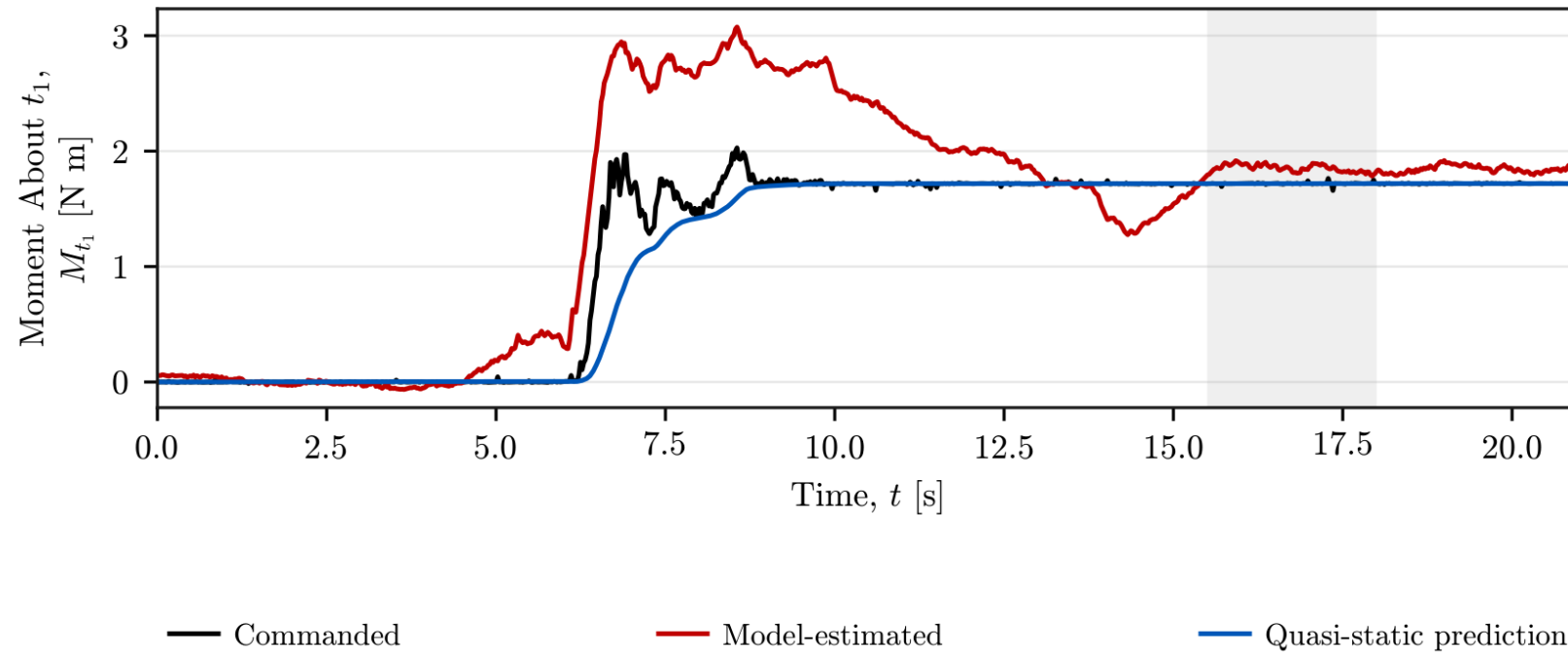
- Stationary force difference: 2.32%
- Stationary moment difference: 4.87%
- Supplementary plausibility evidence

$$F_{n,qs} = K_{p,n} n_s^\top (e_p - e_{p,0})$$



- Spring vs command: 0.05% difference
- Estimate vs command: 13.50% difference

$$M_{t_1,qs} = K_{R,t_1} t_1^\top (e_R - e_{R,0})$$



- Spring and commanded mean: 1.719 N m
- Model-estimated mean: 1.852 N m

The EE frame is attached to the controlled tool centre point (TCP).

Position

$$p_{EE} \in \mathbb{R}^3$$

3 translational degrees of freedom
TCP position in the robot base frame

Orientation

$$R_{EE} \in SO(3)$$

3 rotational degrees of freedom
EE rotation relative to the robot base

Measured pose

$$T_{EE} = \begin{bmatrix} R_{EE} & p_{EE} \\ 0^\top & 1 \end{bmatrix}$$

Obtained directly from the robot state for position and orientation feedback.

7 joint coordinates control a 6-DOF tool pose: 3 position + 3 orientation.

$$\dim(\ker J) = 7 - \text{rank}(J) = 7 - 6 = 1$$

At a nonsingular configuration: one independent redundant motion direction.

This freedom is a coordinated motion of the joints, not one designated joint.

$$J(q) \dot{q}_{\text{null}} = 0, \quad J \in \mathbb{R}^{6 \times 7}$$

J maps joint velocity to tool velocity: null-space motion gives zero tool velocity.

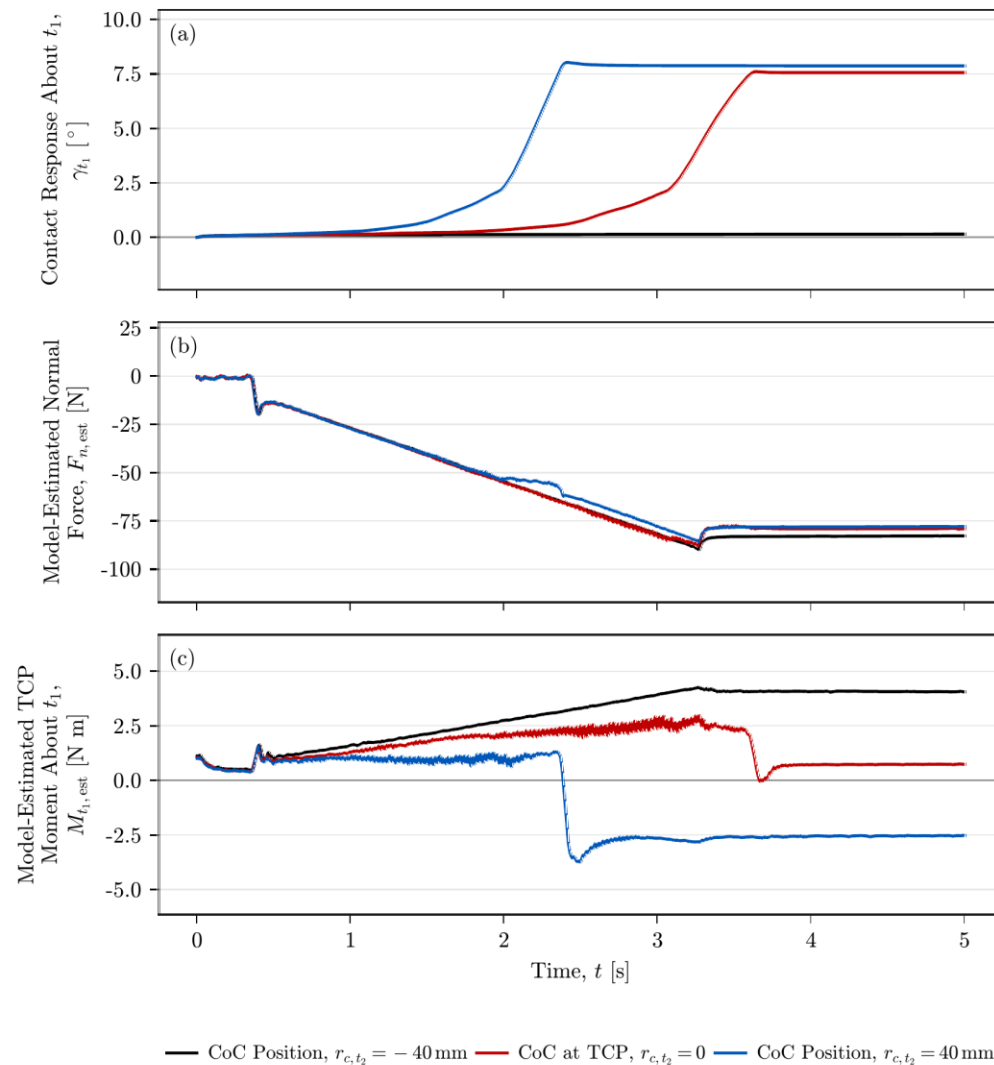
Two torque components control this internal motion:

$$\tau_{\text{null}} = \tau_d + \tau_{\sigma}$$

Damping torque: counters null-space joint motion.

Conditioning torque: steers towards a configuration with better motion capability.

Contact response, normal force and moment



Representative traces Positive entry tilt

- Similar final normal force
- Moment reverses with CoC displacement
- Force and moment: model-based robot estimates